

Computer Organization and Architecture

Lab Performance (50 Marks)

Time: 1 hour 20 minutes

1. Write a program to
 - A. display a string "Input" as a prompt to enter two numbers
 - B. read two numbers from the user and find the sum,
 - C. display the numbers and their sum on the next line with an appropriate message that shows whether the sum is greater than 12 or not.
2. Write a program to
 - A. prompt "ENTER THREE INITIALS:" and read three uppercase letters one by one,
 - B. validate the input and show an error message for invalid input
 - C. display them in separate lines with line no in alphabetically reverse order.
3. Write a program to take input of 3 numbers and show which number is the smallest and which number is the largest.

<i>Sample Input:</i> 1 3 5	<i>Sample Output:</i> 1 is smallest 5 is largest
-------------------------------------	--

4. Write a program to (a) read three characters (c) display them down the left margin in reverse order.

Sample Output:

ENTER THREE INITIALS: ABC

C

B

A

5. Trace the output of the following:

.MODEL SMALL	JL P1	JMP P0	MOV AH,4CH
.STACK 100H	JE P2		INT 21H
.DATA	JG exit	PRINT:	MAIN ENDP
PH DB 1	P1:	MOV AH,9	END MAIN
XH DB 33H	ADD BH,30H	LEA DX,HMSG	
ZH DB 35H	MOV AH,2	INT 21H	
HMSG DB " Streets \$"	MOV DL,BH		
BMSG DB "No Streets\$"	INT 21H	MOV BH,30H	
NEWLINE DB 10,13,'\$'	INC CH	MOV AL,ZH	
	ADD PH,2	MOV XH,AL	
.CODE	JMP P2		
MAIN PROC	P2:	JMP P2	
MOV AX,@DATA	CMP BH,XH		
MOV DS,AX	JE PRINT	exit:	
MOV CH,PH	MOV AH,9	MOV AH,9	
MOV CL,8D	LEA DX,NEWLINE	LEA DX,BMSG	
P0:	INT 21H	INT 21H	
MOV BH,PH	INC CH	MOV AH,2	
CMP CH,CL		MOV DL,CH	
		INT 21H	

6. Trace the output of the following code.

[15]

```
.MODEL SMALL
.STACK 100H
.DATA
    NAMMSG DB "YOUR NAME: MR. JOHN", 0DH, 0AH, '$'
    NUMMSG1 DB "YOUR First NUMBER is: ", '$'
    NUMMSG2 DB "YOUR SECOND NUMBER is: ", '$'
    SUMMSG DB "SUM OF DIGITS: ", '$'
    SUBMSG DB "SUB OF DIGITS: ", '$'
    NEWLINE DB 0DH, 0AH, '$'
    DIGIT1 DB 5
    DIGIT2 DB 4
.CODE
MAIN PROC
    MOV AX, @DATA
    MOV DS, AX
    LEA DX, NAMMSG
    MOV AH, 09H
    INT 21H
    LEA DX, NUMMSG1
    MOV AH, 09H
    INT 21H
    MOV AL, DIGIT1
    ADD AL, '0'
    MOV DL, AL
    MOV AH, 02H
    INT 21H
```

```
    LEA DX, NEWLINE
    MOV AH, 09H
    INT 21H
    LEA DX, NUMMSG2
    MOV AH, 09H
    INT 21H

    MOV AL, DIGIT2
    ADD AL, '0'
    MOV DL, AL
    MOV AH, 02H
    INT 21H
```

```
    LEA DX, NEWLINE
    MOV AH, 09H
    INT 21H
```

```
    LEA DX, SUMMSG
    MOV AH, 09H
    INT 21H
    MOV AL, DIGIT1
    ADD AL, DIGIT2
    ADD AL, '0'
    MOV DL, AL
    MOV AH, 02H
    INT 21H
```

```
    LEA DX, NEWLINE
    MOV AH, 09H
    INT 21H
    LEA DX, SUBMSG
    MOV AH, 09H
    INT 21H
    MOV AL, DIGIT1
    SUB AL, DIGIT2
    ADD AL, '0'
    MOV DL, AL
    MOV AH, 02H
    INT 21H

    MOV AH, 4CH
    INT 21H
MAIN ENDP
END MAIN
```